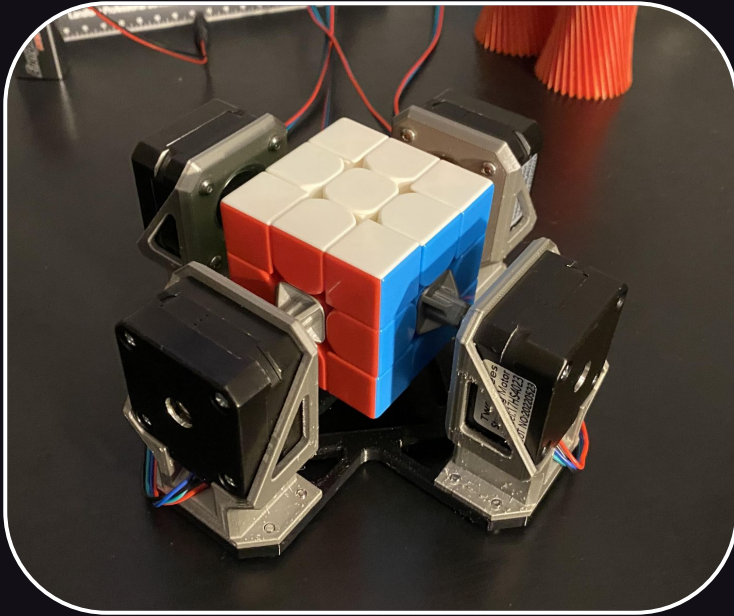
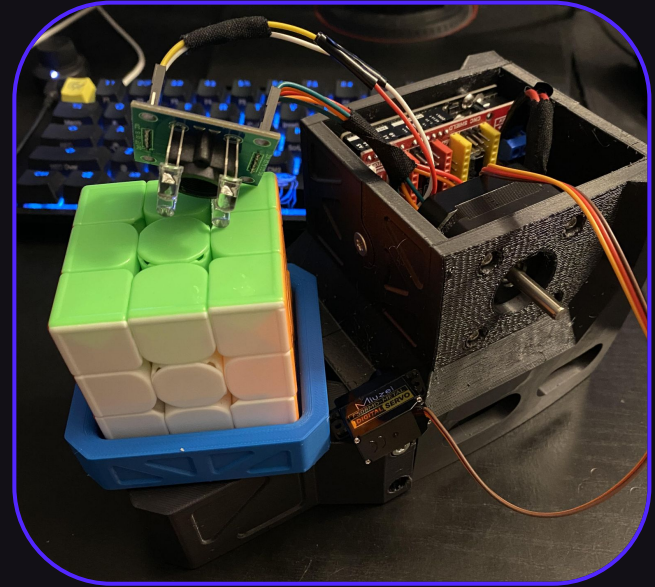


Original Designs

I originally solved this project with a 4 stepper solution and arduino. This worked, but the solution to the cube had to be input manually.

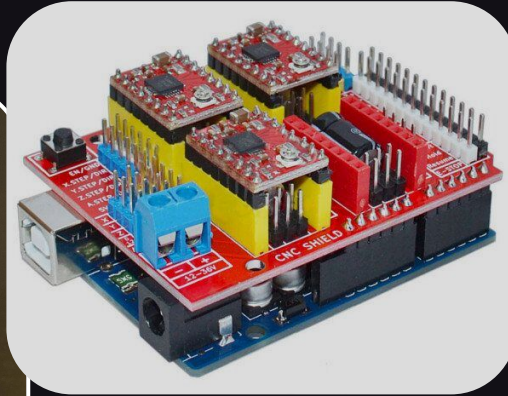
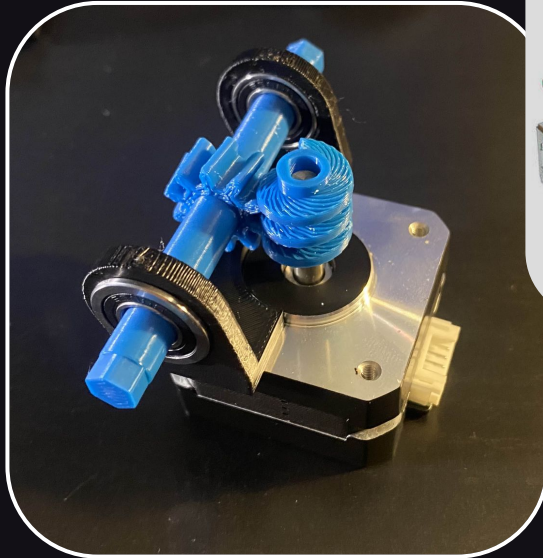


Next, I made a design that had a color sensor to scan each face. This design also worked, but the component that flipped the cube could not be moved by the small stepper motor.



Final Design

To make lifting the color sensor and servo easier, I used a worm gear to gain mechanical advantage. In this design, the cube isn't locked into the mechanism and the user can place it.



I used the arduino cnc shield attachment to drive the 2 stepper motors in the assembly. All of the wiring including the servo, color sensor and steppers is contained within the assembly.